

Distances & Divergences — Quick-Reference Table

Grouped families, defining equations, and typical applications. Generator convention: f -divergence $D_f(p\|q) = \int p f(q/p) d\mu$; Bregman B_F from convex F . $\parallel$ marks asymmetric (divergence) arguments; argument-order conventions vary by source.

Group	Distance / divergence	Equation	Application — where to use
Vector & metric distances	Euclidean (L_2)	$d_2 = \sqrt{\sum_i (p_i - q_i)^2}$	k -means, k -NN, least squares
	Manhattan (L_1)	$d_1 = \sum_i p_i - q_i $	Grid/route cost; robust to outliers; LASSO
	Minkowski (L_k)	$d_k = (\sum_i p_i - q_i ^k)^{1/k}$	Tunable; unifies L_1, L_2, L_∞
	Chebyshev (L_∞)	$d_\infty = \max_i p_i - q_i $	Max-coordinate gap; chessboard moves
	Mahalanobis	$\sqrt{(p - q)^\top \Sigma^{-1} (p - q)}$	Scale/correlation-aware; outliers, classification
	Quadratic	$\sqrt{(p - q)^\top Q (p - q)}$	Feature-weighted; colour histograms
	Hamming	$ \{i : p_i \neq q_i\} $	Error-correcting codes, DNA, bit strings
Riemannian & information geometry	Fisher information	$\mathbf{I}(\theta) = \mathbb{E}[(\partial_\theta \ln p)(\partial_\theta \ln p)^\top]$	Natural metric on models; Cramér–Rao, natural gradient
	Fisher–Rao distance	$\rho = \min_\gamma \int_0^1 \sqrt{\dot{\gamma}^\top \mathbf{I} \dot{\gamma}} dt$	Intrinsic geodesic distance between distributions
	Riemannian geodesic	$L = \int \sqrt{g_{ij} \dot{x}^i \dot{x}^j} dt$	Shortest path on curved manifolds; shape spaces
f-divergences (Csiszár / Ali–Silvey)	<i>general template</i>	$D_f(p\ q) = \int p f(q/p) d\mu$	Master family — pick the convex generator f
	Kullback–Leibler	$\int p \log \frac{p}{q} d\mu$	MLE, cross-entropy loss, variational inference
	Pearson χ^2	$\int \frac{(q-p)^2}{p} d\mu$	Goodness-of-fit; local KL approximation
	Hellinger	$\sqrt{\frac{1}{2} \int (\sqrt{p} - \sqrt{q})^2 d\mu}$	Symmetric bounded <i>metric</i> ; robust statistics
	Total variation	$\frac{1}{2} \int p - q d\mu$	Statistical distinguishability; coupling
	Amari α -divergence	$f_\alpha(t) = \frac{4}{1-\alpha^2} (1 - t^{\frac{1+\alpha}{2}})$	One dial: KL ($\alpha=1$), Hellinger ($\alpha=0$)
Bregman divergences	<i>general template</i>	$B_F(x\ y) = F(x) - F(y) - \langle x - y, \nabla F(y) \rangle$	Master family — pick the convex potential F
	Squared Euclidean	$\ x - y\ ^2$	Centroid clustering (k -means)
	Itakura–Saito	$\sum_i (\frac{p_i}{q_i} - \log \frac{p_i}{q_i} - 1)$	Audio/spectral distortion; NMF
	Log-Det	$\langle P, Q^{-1} \rangle - \log \det(PQ^{-1}) - n$	SPD matrices; metric learning (ITML)

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Overlap / α-power family	Bhattacharyya	$-\log \int \sqrt{pq} \, d\mu$	Class separability; object tracking
	Chernoff information Rényi divergence	$\max_{\alpha \in (0,1)} -\log \int p^\alpha q^{1-\alpha} d\mu$ $\frac{1}{\alpha-1} \log \int p^\alpha q^{1-\alpha} d\mu$	Error exponent in hypothesis testing Differential privacy; information theory
Symmetrized & Jensen-type	Jeffreys	$\text{KL}(p\ q) + \text{KL}(q\ p)$	Symmetric KL when direction is arbitrary
	Jensen–Shannon Burbea–Rao / Jensen	$\frac{1}{2} \text{KL}(p\ m) + \frac{1}{2} \text{KL}(q\ m), \, m = \frac{p+q}{2}$ $\frac{F(p)+F(q)}{2} - F\left(\frac{p+q}{2}\right)$	Bounded; $\sqrt{\cdot}$ is a metric; GANs Jensen gap; JS is the Shannon-entropy case
Entropies (functionals behind divergences)	Shannon / Boltzmann	$H = -\int p \log p \, d\mu$	Information content, source coding
	Rényi	$H_\alpha = \frac{1}{1-\alpha} \log \int p^\alpha d\mu$	Min-/collision-entropy; cryptography
	Tsallis (non-additive)	$T_\alpha = \frac{1}{1-\alpha} (\int p^\alpha d\mu - 1)$	Non-extensive (long-range) systems
	Von Neumann (quantum)	$S(\rho) = -\text{Tr}(\rho \log \rho)$	Quantum entropy; entanglement
Set & metric-space distances	Hausdorff	$\max\{\sup_x \rho(x, Y), \sup_y \rho(X, y)\}$	Set/shape matching; image comparison
	Gromov–Hausdorff	$\inf_{\phi_X, \phi_Y} \rho_H^Z(\phi_X X, \phi_Y Y)$	Compare spaces up to isometry; manifold learning
Optimal transport & IPMs	Wasserstein / EMD	$(\inf_{\gamma \in \Gamma} \int \rho(x, y)^\alpha d\gamma)^{1/\alpha}$	Optimal transport; WGAN, retrieval; EMD= W_1
	Max Mean Discrepancy	$\sup_{\ f\ _{\mathcal{H}} \leq 1} \mathbb{E}_p f - \mathbb{E}_q f $	Kernel two-sample tests; generative models
	Kolmogorov–Smirnov	$\sup_x F_p(x) - F_q(x) $	Non-parametric goodness-of-fit
	Lévy–Prokhorov	$\inf\{\varepsilon : p(A) \leq q(A^\varepsilon) + \varepsilon \, \forall A\}$	Mettrizes weak convergence (in distribution)
Quantum geometry	Von Neumann divergence	$\text{Tr}(P(\log P - \log Q) - P + Q)$	Quantum KL; distinguishing quantum states

Nesting at a glance: $L_1 \subset L_2 \subset L_\infty$ are Minkowski cases; Mahalanobis is Quadratic with $Q = \Sigma^{-1}$. KL is the *only* divergence that is both an f -divergence *and* a Bregman divergence. Bhattacharyya/Chernoff/Rényi all come from the affinity $\int p^\alpha q^{1-\alpha} d\mu$. Wasserstein, MMD, KS, Lévy–Prokhorov are IPMs. Reorganized from Frank Nielsen’s taxonomy chart.